6 **(a)** A nuclide is radioactive if it is unstable and may undergo random and spontaneous disintegration, and structural changes within its nucleus, to transform into another nuclide.

In the process, radiation may be emitted from the nucleus in the form of either alpha particles, beta particles, gamma radiation or a combination of these.

Note: Must explain that it is the nucleus that emits the radiation.

(b) **(i)** Decay constant is defined as the probability of a radioactive nucleus disintegrating per unit time interval.

(ii) Number of atoms in 2.40×10^{-8} of Strontium-90 is

$$N=\frac{2.40\times10^{-8}}{90}(6.02\times10^{23})\approx1.6053\times10^{14}$$

Activity $A = \lambda N$

Decay constant $\lambda = \dfrac{A}{N} = \dfrac{1.26 \times 10^{5}}{1.6053 \times 10^{14}} = 7.85 \times 10^{-10}\ \text{s}^{-1}$

(c) The activity in radioactivity decay is not a constant value, but decreases with time as the number of radioactive nuclide remaining falls.

However, if the decay constant is small, then the half-life is long ($t_{\frac{1}{2}} = \dfrac{\ln 2}{\lambda}$). Hence over a relatively short time interval during which the activity is measured (1 hour in this question), the activity remains fairly stable, and can be assumed to be constant.